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PREVIEW

GRADES 7–10

EMERGENCY SUB PLANS PACK

for DIGITAL TECHNOLOGIES

★ *Keep learning on track—even when plans change.* ★

- **NO-PREP ACTIVITIES**
- **READY-TO-USE RESOURCES**
- **PERFECT FOR EMERGENCY LESSONS**
- **GREAT FOR SUBSTITUTE TEACHERS**
- **ENGAGING DIGITAL TECH TASKS**
- **PRINT OR DIGITAL USE**

YOUR ***LIFESAVER*** IN THE CLASSROOM

SAVE TIME | REDUCE STRESS | KEEP STUDENTS ON TRACK

MADE BY TEACHERS FOR TEACHERS

BEFORE YOU NEED IT

Set this up while you are well

FOR THE REGULAR TEACHER

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a day when you are not.

The one job you have today

Print pages 3–6 of this pack and leave them in a clearly labeled folder where your substitute teacher will find them — top drawer, on the desk, or with the front office. Everything a substitute needs is in those four pages.

You do not need to print the lessons in advance. The substitute teacher chooses one on the day and photocopies it.

SETTING IT UP · FIFTEEN MINUTES, ONCE

1. **Print the whole pack once.** Put it in a display folder or bind it. Write your name and room number on the cover.
2. **Print pages 3–6 a second time** and clip them to the front. These are the pages a substitute teacher reads first.
3. **Tell your department head where it lives.** On a genuine emergency day you will not be answering your phone, so somebody else needs to know.
4. **Cross off the lessons you have already used.** There is a tracker on page 4. Two years of substitute days will not repeat if you keep it current.
5. **Optional: leave a class list** and a note about any students who need specific support, seating or a reader.

Why these lessons are unplugged

On a substitute day the laptop cart is often locked, the login server is down, or nobody knows the room password. Every lesson here runs on paper.

Each lesson also has an **If the computers are working** note, so the same sheet still works when technology cooperates.

Why they are not busy work

Every lesson targets a genuine Digital Technologies concept and produces marked evidence you can put in a folder.

If you are away for a week, running four of these in sequence is a defensible unit of work, not a holding pattern.

A word about the answer keys

Every answer key sits on the **last page of its lesson**. If you hand a lesson to a substitute teacher, hand them the answer key too — it is the single thing that lets a non-specialist mark with confidence and answer a student who says *“but why?”*

This is Volume One

Volume 1 is the foundation pack: algorithms, binary, tracing, flowcharts, security, networks, efficiency, debugging, AI and privacy. Volumes 2, 3 and 4 add thirty further lessons and none of them repeat anything here.

SUBSTITUTE TEACHER · START HERE

You do not need to know this subject

READ THIS FIRST

2 MINUTES

You have walked into a Digital Technologies class. You may have never taught the subject. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to know how to code. Every lesson is a paper worksheet with a full answer key. Your job is to hand it out, keep the room settled, and check work as students finish. You are not expected to teach the content.

DO THESE FOUR THINGS

1. **Pick one lesson** from the table on the next page. Any of them works for any class from Grade 7 to Grade 10. If you cannot decide, use Lesson 05.
2. **Photocopy the student pages only.** Each lesson tells you exactly which page numbers to copy. One copy per student.
3. **Keep the teacher page and the answer key.** Do not photocopy those.
4. **Read the teacher page once** before the bell. It takes about two minutes and includes the exact words you can say to start the lesson.

If a student says “we have done this”

Ask them to prove it by completing the extension section at the end. If they genuinely have, switch them to a different lesson from the table — the pack has ten, and they will not have done them all.

If you finish early

Every lesson has an extension task on its final student page. Point students there. There is also a discussion prompt on the teacher page you can run as a whole class for the last five minutes.

What to leave for the regular teacher

Collect the completed sheets. Fill in the handover note on the last page of this pack — it takes ninety seconds and tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

SUBSTITUTE TEACHER · STEP ONE

Choose your lesson

PICK ANY ONE

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	Follow My Instructions Exactly	Algorithms and precision	★	60	7–9
02	Counting Like a Computer	Binary and data representation	★★	60	12–14
03	Trace the Code	Reading programs by hand	★★	55	17–19
04	Flowchart Detective	Logic and decisions	★	55	22–24
05	How Attacks Actually Work	Cyber security	★	60	27–29
06	Sending a Message Across the World	Networks and the internet	★★	60	32–34
07	Sorting and Searching by Hand	Algorithm efficiency	★★★	60	37–39
08	Bug Hunt	Debugging and error types	★★★	55	42–44
09	What AI Can and Cannot Do	Artificial intelligence and ethics	★	60	47–49
10	Your Digital Footprint	Privacy and digital citizenship	★	55	52–54

Easiest to supervise

05, 09 and 10. Scenario and discussion based. Students have opinions, so the room stays busy and nobody gets stuck.

Best for a quiet, working class

02, 03 and 07. Puzzle-like with definite answers. Strong students will race; that is what the extensions are for.

Best for a restless class

01 and 04. Partner activities with movement and a bit of humor built in. Noisier, but productively so.

Used-lesson tracker · for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Instructions			
02 — Binary			
03 — Trace the code			
04 — Flowcharts			
05 — Cyber security			
06 — Networks			
07 — Sorting			
08 — Bug Hunt			
09 — AI			
10 — Digital Footprint			

SUBSTITUTE TEACHER · STEP TWO

The first ten minutes

WHAT TO SAY

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

> *"Come in, sit down, and leave your bags off the desks. You will not need a computer today. Everything is on paper."*

THEN, ONCE THEY ARE SEATED

> *"Your teacher is away. My name is ______. We are doing a proper Digital Technologies lesson today, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."*

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their regular teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Settle, hand out sheets	Use the script above
5–10	Read the opening page together	Ask one student to read it aloud
10–45	Students work through the tasks	Circulate. Check answers against the key
45–55	Extension task for those who finish	Point them to the last student page
55–60	Collect sheets, pack up	Fill in the handover note

If a student refuses to work

Offer the smaller ask: *"Just do question one and show me."* Most refusals are about not knowing where to start, not defiance.

Note it on the handover sheet. Do not escalate a single lesson into a battle you will not be there to finish.

If the whole class claims they cannot do it

Do the first question together on the board. Every lesson has a worked example on the first student page.

If it is genuinely too hard for the grade level, switch to Lesson 05 or Lesson 10 — both need no prior knowledge at all.

Photocopying reminder

Student pages only. Each lesson's teacher page states the exact page range. The **answer key is always the last page** of the lesson — keep it.

LESSON 01 · TEACHER BRIEFING

Follow My Instructions Exactly

READ BEFORE THE BELL

2 MINUTES

Students discover that a computer does precisely what it is told — which is rarely what you meant. It is the foundation of every algorithm they will ever write, and it is genuinely funny when it goes wrong.

You do not need to know this subject

This lesson is about writing clear instructions in plain English. There is no code and no technical vocabulary you need to hold. If you can tell whether a sentence is vague, you can run this lesson.

Photocopy these pages only

Pages 7–9

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

> *"Today you are going to find out how annoying a computer actually is. It will do exactly what you say. Not what you meant — what you said. By the end you will understand why programmers spend so much time being precise."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the opening panel together	Have a student read 'the obedient idiot' aloud
12–22	Tasks 1.1 and 1.2 — spotting vagueness	Circulate. Task 1.1 has no single right drawing
22–38	Task 1.3 — write and swap the sandwich algorithm	This is the noisy part. Let it be noisy
38–50	Tasks 1.4 and 1.5 — loops and decisions	Answer key has model answers
50–60	Collect sheets, run the discussion prompt	Prompt is at the foot of this page

Questions students will ask

"Is my drawing wrong?"
No. That is the point of Task 1.1. The instructions do not say how big or which direction, so several different drawings are all correct.

"How many steps should the sandwich have?"
At least eight. If a student writes four, ask them what a robot would do with step one.

"Can I write 'spread the peanut butter'?"
Ask: with what? Which side? How much? That question is the whole lesson.

If they finish early

Task 1.5 on the third student page is the extension — writing instructions for a robot that has to make a decision and repeat itself. Strong students can then write instructions for tying a shoelace, which is far harder than it sounds.

If the computers are working

Students can type their sandwich algorithm and have a classmate follow it literally on screen, or try it in any block-based coding tool where the computer really will refuse a vague instruction.

Curriculum focus · for the regular teacher's records

Designing algorithms; representing a solution as a sequence of steps; using branching and iteration in a plain-language algorithm. Discussion prompt for the last five minutes: *"A human can follow 'add a little salt'. Why can a computer never follow it — and is that a weakness or a strength?"*

The obedient idiot

UNDERSTAND

12 MIN

NAME

CLASS

DATE

Computers are not clever. They are fast and obedient.

A computer will carry out your instructions perfectly, in order, without ever stopping to ask what you meant. That sounds helpful. In practice it means a single vague word can ruin an entire program.

Humans fill in gaps automatically. If someone says “grab me a drink” you work out the rest. A computer has no idea what “a drink” is, where it is, or what “grab” means.

Task 1.1 · Draw exactly what it says

DO

Follow these five instructions in the box. Do not use common sense. Do only what the words actually say.

instructions — FIVE STEPS

1. Draw a dot near the middle of the box.
2. Draw a straight line starting at the dot.
3. From the end of that line, draw another line.
4. Join the end of the second line back to the dot.
5. Shade in the shape you have made.

Now compare with the person next to you. Your drawings are almost certainly different. List **four** things the instructions never told you:

LESSON 01 · TEACHER COPY

Answer key

Follow My Instructions Exactly

NOT FOR STUDENTS

ANSWERS

How to grade this lesson

Most answers here are **judgment calls, not exact matches**. Give the credit if the student has removed the ambiguity, even if their wording differs from the model answer below.

Task 1.1 · what the instructions never said

Any four of: how big the dot is · how long each line is · in which direction each line goes · whether the lines are straight · what "near the middle" means · what color to shade · whether the shape is closed at all.

All differing drawings are correct — that is the point.

Task 1.2 · vague words

#	VAGUE PART	MODEL REWRITE
a	a little	Add 1/2 teaspoon of salt.
b	it	Move the pointer over the OK button, then press the left mouse button once.
c	sort	Arrange the list from smallest to largest.
d	a while	Wait 30 seconds, then try again.
e	looks right	Repeat until the number reaches 100.

Task 1.3 · sandwich algorithm

Full credit for eight or more steps that each contain a single action. Look for: which item is picked up, which surface, which side, how much, and what to do with the knife. Almost every class forgets to **open the jar** and to **take the bread out of the bag**. Those are the best teaching moments — point them out rather than grading them wrong.

Task 1.4 · model answer

after — PLAIN ENGLISH

```
REPEAT 6 TIMES:
    water the next plant
```

Accept any version that repeats once and moves to the next plant each time.

Task 1.5 · model answer

seven days — PLAIN ENGLISH

```
REPEAT 7 TIMES:
    check the soil
    IF the soil is dry:
        pour 1 cup of water on it
    OTHERWISE:
        do nothing
    wait 24 hours
```

The point of the last question

A computer's refusal to guess is what makes it **reliable and repeatable**. A human cook adds a different amount of salt every time; a machine adds exactly half a teaspoon, a million times, identically. Accept any answer that reaches consistency, reliability or repeatability.

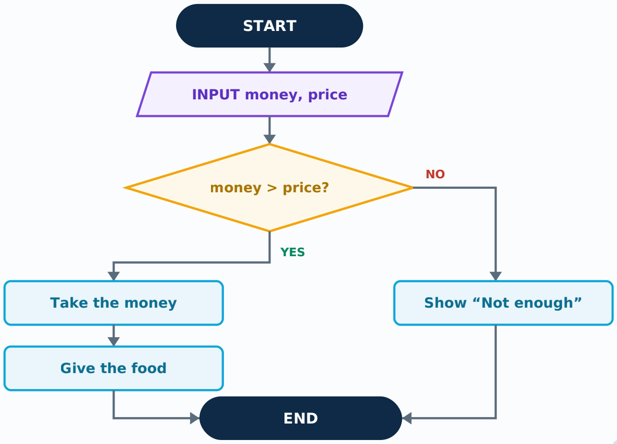
Find the bug that costs money

KEY SKILL
20 MIN

Task 4.2 · The school cafeteria

DETECTIVE

This flowchart runs the self-serve register at a school cafeteria. It works nearly all the time — but there is a logic error in it that quietly loses the cafeteria money.



First, follow it normally:

MONEY	PRICE	WHAT HAPPENS
$5.00	$3.50	
$2.00	$3.50	
$3.50	**$3.50**	

a. Which row above shows the problem?

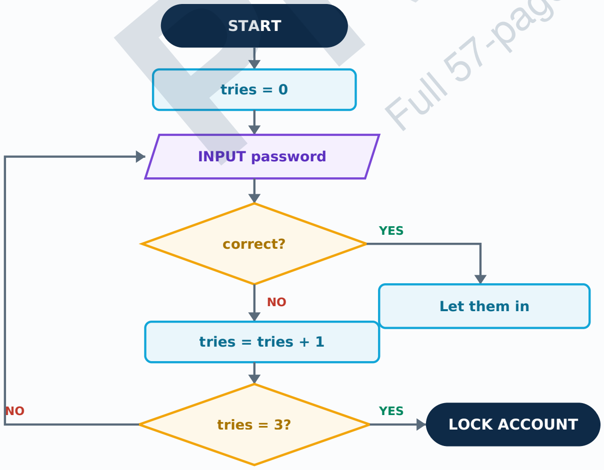
b. Describe what goes wrong in that case:

c. Write the one small change that fixes it:

Task 4.3 · The login screen

TRACE

This one controls a login screen. Notice the arrow that goes **backwards** — that is a loop.



a. How many times can someone get the password wrong before the account locks?

b. Which box makes the loop possible?

c. What would happen if the box `tries = tries + 1` were deleted?

d. Why is it safer to lock the account than to keep asking forever?

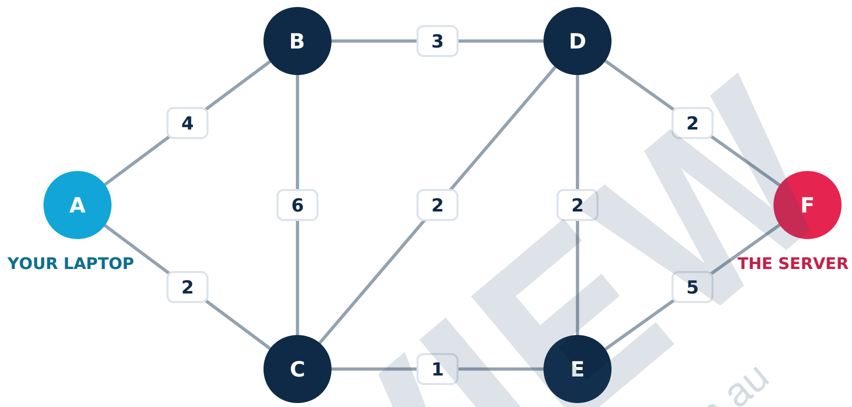
Finding the way

KEY SKILL

18 MIN

Every link has a cost

Each line below is a cable between two routers. The number is how long that hop takes. Your laptop is **A**. The server you are trying to reach is **F**.



Task 6.3 · Find the cheapest route

CALCULATE

Add the numbers along each route. The cheapest total wins.

ROUTE	WORKING	TOTAL
A → B → D → F		
A → C → E → F		
A → C → D → F		
A → B → C → E → F		

a. The cheapest route is ______________________ with a total of ______

b. The cable between C and E breaks. Which route would the network use instead, and what does it cost now? __________________________________

AFTER THE LESSON

Handover note

LEAVE THIS ON THE DESK

90 SECONDS

Photocopy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

SUBSTITUTE TEACHER	CLASS	DATE	PERIOD

Lesson used

CHECK ONE

- ☐ 01 · Follow My Instructions Exactly
- ☐ 02 · Counting Like a Computer
- ☐ 03 · Trace the Code
- ☐ 04 · Flowchart Detective
- ☐ 05 · How Attacks Actually Work
- ☐ 06 · Sending a Message Across the World
- ☐ 07 · Sorting and Searching by Hand
- ☐ 08 · Bug Hunt
- ☐ 09 · What AI Can and Cannot Do
- ☐ 10 · Your Digital Footprint

How far did the class get?

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

Anything the regular teacher should know

NOTES

Students to mention · for any reason, good or otherwise

NAMES

Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is the part that makes the next teacher’s day work, and it is genuinely appreciated.

THE FINE PRINT

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DIGITAL VECTOR

VOLUME ONE

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Curriculum alignment

CSTA K–12 Computer Science Standards, levels 2 and 3A. Across the ten lessons this volume touches:

- Designing and representing algorithms
- Representing data in binary and other forms
- Tracing and desk-checking programs
- Branching, iteration and logic errors
- Cyber security, passwords and social engineering
- How networks transmit and route data
- Algorithm efficiency: sorting and searching
- Testing, debugging and error types
- Artificial intelligence, evidence and ethics
- Digital footprints, privacy and personal data

Individual lesson teacher pages name the specific focus for that lesson.

The rest of the series

Volume 2 · Data, Systems and Design

Data cleaning, misleading charts, databases, input-process-output, compression, encryption, interface design, functions, project planning and the cost of free apps.

Volume 3 · Machines and Meaning

Lists, Boolean logic, hardware, files, requirements, machine learning, DNS, search ranking, control systems and copyright.

Volume 4 · Media, Evidence and Impact

Images and sound as data, stacks and queues, digital forensics, version control, network design, accessibility, algorithmic decisions, IoT security and automation.

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Physical Computing with the Raspberry Pi 5 · AI Literacy for Secondary Students · and more.

Found an error, or want a lesson on something else?

Feedback is genuinely read. If a lesson did not land with your class, or you need a sub plan on a topic that is not here, get in touch through digitalvector.com.au and it will be considered for the next update.

EMERGENCY SUB PLANS PACK

Digital Technologies · Grades 7–10

Volume One · ten lessons · no preparation · no computers required